

SACHET RANJAN BISI

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Education

New York University

May 2025

Master of Science in Computer Engineering

New York City, NY

- Relevant Coursework: Machine Learning, Artificial Intelligence, Internet Protocols, Network Security

University of Illinois

May 2023

Bachelor of Science in Computer Engineering

Chicago, IL

- Relevant Coursework: Data Structures, Computer Vision, Systems Programming, Computer Architecture

Technical Skills

Languages: Python, Java, JavaScript, SQL, C++

Frameworks & Libraries: FastAPI, Django REST, TensorFlow, PyTorch, Scikit-learn, Keras, LangChain, RAG

Cloud & DevOps: AWS, GCP, Docker, Jenkins, GitLab CI/CD, Kubernetes

Web Development: React.js, RESTful APIs, Microservices

Tools: VS Code, Git, Postman, Jupyter Notebook, PyTest, JUnit

Experience

Viabot

May 2024 – August 2024

Software Engineer Intern

Remote

- Developed and deployed scalable microservices using Python (FastAPI) on AWS and GCP, improving system uptime by 40% and enabling faster cross-platform integrations.
- Automated deployment pipelines with Jenkins and GitLab CI/CD, improving deployment consistency and reducing manual intervention by 30%, while cutting average release time and accelerating feature delivery.
- Refactored server-side components in Spring Boot, optimizing resource utilization and boosting backend throughput by 60%.
- Built high-performance RESTful APIs to streamline data exchange between services, cutting integration time by 40%.
- Strengthened reliability through extensive unit and integration testing (PyTest, JUnit), achieving 95% coverage and lowering post-release defect rates by 35%.

University of Illinois at Chicago

May 2022 – December 2022

Computer Engineering Intern

Chicago, IL

- Collaborated on the development of an IR thermal camera prototype, integrating embedded hardware and Python-based signal processing to enhance temperature detection accuracy.
- Automated circuit validation workflows and optimized component wiring using Fritzing, reducing manual debugging by 25%.
- Diagnosed and resolved key circuit design issues, improving prototype reliability and performance consistency by 30%.

Projects

LLM-Powered Medication Verification

October 2024

- Designed a multi-modal RAG pipeline integrating OCR APIs and AWS Transcribe, improving text extraction accuracy by 25% for handwritten and audio clinical inputs.
- Implemented intelligent reconciliation workflows using LangChain agentic orchestration with prompt-driven task routing and memory management, cutting medication validation time by 50% and improving LLM response accuracy.
- Developed scalable backend APIs with FastAPI and Django REST to handle asynchronous data ingestion and structured exchange between modules, improving response latency by 60% and enhancing system reliability.

Food Classification Deployment Track

December 2023

- Trained and fine-tuned a CNN (MobileNetV2) model achieving 88% accuracy, containerized it with Docker, and deployed it as a FastAPI microservice on GCP Cloud Run for scalable real-time inference.
- Containerized workloads using Docker and automated CI/CD through Jenkins, reducing model deployment time by 45%.
- Enhanced scalability and latency using Kubernetes orchestration, improving inference performance by 85%.

Beep Baseball

May 2023

- Researched and authored the project proposal, contributing to a team that secured \$250 in funding for the project.
- Conducted rigorous testing on electrical components and collaborated with engineers to identify and resolve critical issues, reducing potential production errors by 30% and enhancing product reliability.

Neural Network Application

December 2022

- Implemented a perceptron training algorithm for image classification, achieving 92% accuracy on validation datasets.
- Optimized curve fitting using backpropagation, reducing error by 86%, and built a Python-based shape classification model, improving efficiency by 89%.